# Yu-Lin (Willy) Liu

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## HIGHLIGHTS OF ACHIEVEMENTS AND QUALIFICATIONS

**Industry Experience:** Game Engine Development Engineer with over 3 years of experience
**Education:** Bachelor of Engineering in Software Engineering
**Programming:** C++, C#, Lua, Java, and Python
**Engines & Tools:** Unity, Jenkins, Spring Boot, Git, Perforce, and PyTorch
**Specialization:** Optimization for open-world, multi-platform, online games with millions of players

- Memory Optimization, Multithreading, and Animation Optimization
- Resource Loading Acceleration
- Custom Data Structure Design
- Developer Tools & Plugin Development

## EDUCATION

**University of Wisconsin-Madison** 09/2026 – 05/2028 | Wisconsin, United States
*Master of Science, Computer Science*

- **Coursework:** Advanced Computer Networks; Advanced Computer Architecture I; High Performance Computing for Applications in Engineering

**Nanjing University** 09/2018 – 06/2022 | Nanjing, China
*Bachelor of Engineering, Software Engineering*

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Operating System, Software Engineering, Software System Design, Machine Learning, Pattern Recognition

## WORK EXPERIENCE

**Game Engine Development Engineer** 07/2022 – 04/2025 | Shanghai, China
*Lilith Games, AFK: Journey*

- Developed game systems in Unity using Lua, C++, C#, and Python.
- The game received multiple awards, including App Store Best Game of 2024, Google Play Best Game of 2024, and a nomination for Best Mobile Game at The Game Awards 2024.

**Memory Optimization**

- Reduced memory usage from 2.5GB to 1GB by eliminating redundant allocations, designing custom data structures, and using memory-mapped loading.
- Resolved memory leaks, enabling iPhones with 3GB RAM to run the game for over 1 hour by identifying zombie memory, mitigating memory fragmentation, and reducing frequent allocations.

**3D Character Production Pipeline**

- Compressed animation memory usage by removing frames with only subtle changes, reducing asset memory footprint from 10MB to 1MB.
- Assembled all artist resources into complete character assets.
- Implemented automated tools for animation clipping detection and batch modification of model vertex normals.
- Designed an animation play system to dynamically load clips and controllers, reducing animation memory usage from 100MB to 15MB.

**Loading Acceleration**

- Accelerated game startup loading time from 40 to 10 seconds by simplifying the parsing of map data.
- Designed asynchronous resource loading strategies to minimize waiting times during system transitions.

**Open-world Map Optimization**

- Rendered over 500 complex 2D icons on screen while allowing players to drag the 2D map at a high, stable frame rate.
- Optimized a level-of-detail (LOD) system for the Infinite Map Zoom System to reduce the number of vertices on screen.
- Designed terrain-adjustment algorithms for curved maps to enhance control fluidity, ensuring natural camera movement during drag interactions.
- Optimized a multithreaded map mesh rendering pipeline by rearranging data read sequences to reduce the number of draw calls.

**Tools for Effect Evaluation**

- Built tools to provide artists with clear standards for checking visual effect performance during development.
- Developed an effect grading system to control and monitor the performance cost of in-game effects.

**Game Engine Development Internship** 07/2021 – 02/2022 | Shanghai, China
*Lilith Games, AFK: Journey*

**Infinite Map Zoom System**
- Optimized the system so the map can be dragged or pinched to the edge without noticeable loading and can transition smoothly from a 3D map to a 2D map.
- Developed a camera movement system to control camera behavior on the map.
- Implemented gesture events (drag and pinch) for the map and icon interaction logic such as touch responses.

**2D Mini-map Rendering System**
- Rendered 2D icons on the mini-map and implemented their rule-based display management system.

**Developer Tools for Designers and Artists**
- Implemented a map-copy tool.
- Built a FlatBuffers-based map data export pipeline.
- Developed a map resource management tool.

## RESEARCH EXPERIENCE

**Research Assistant** 05/2025 – 08/2025 | Kaohsiung, Taiwan
*National Sun Yat-sen University, Mechatronics in Medicine Lab*
- Predicted age from single-lead electrocardiograms (ECG) in polysomnography (PSG) to estimate the apnea–hypopnea index (AHI), an indicator of sleep apnea severity.
- Utilized 1D and 2D machine learning models. Applied ResNet-18, MultiRocket, TabPFN, XGBoost, and other ECG foundation models from recent research.
- Worked with real-world data from a hospital sleep medicine center, requiring extensive noise cleaning and preprocessing.
- Achieved 0.95 training accuracy on real-world data by cleaning noisy data and combining different models to correct errors.

**Graduation Design (Capstone Project)** 03/2022 – 06/2022 | Nanjing, China
*Nanjing University, Software Institute*
- Utilized a 2D-CNN to detect faults in app UIs. Designed a model taking image inputs to generate binary classifications (true/false)**,** achieving 0.87 accuracy on the test dataset.
- Built an automated testing framework to augment the dataset. Collected screenshots from real-world apps and injected faults into normal screenshots to create faulty samples.

## AWARDS

**National Third Prize – Citi Financial Innovation Application Competition** 05/2021
*Nanjing, China*
- **Project development** competition; project name: *T-Bank*.
- Designed a system that enables students to volunteer in remote locations, including an algorithm to match the requirements from remote locations with suitable students.
- Implemented the core backend API, including login verification and user data processing.

**Creative Award – Lilith "Da Vinci Project" Game Creation Competition** 07/2020
*Nanjing, China*
- **Indie game** competition; source code: https://github.com/willy229liu/Lilith-Game-Project-Da-Vinci
- Developed a fully playable 30-minute horror game using Lilith's developed editor, implementing all game logic in Lua.

**National Third Prize – Huawei CodeCraft Competition** 05/2020
*Nanjing, China*
- **Algorithm** competition; source code: https://github.com/1Lucifer1/HuaweiCodeCraft2020
- Optimized C++ code to reduce runtime. Preliminary and semi-final problems involved detecting cycles in directed graphs; the final focused on computing vertex centrality.
- Built custom data structures to avoid STL overhead, tuned code for better cache locality, and redesigned thread management to maximize CPU efficiency.

## SCHOLARSHIPS

- Third Prize, Nanjing University Scholarship for Students from Taiwan (2019, 2020, 2021)
- People's Scholarship for Academic Competitions, Nanjing University (2020)